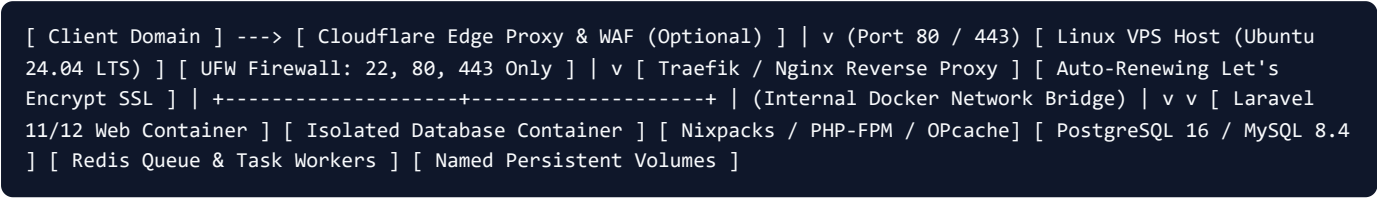


Standard Architecture Directive: Deployed 100% directly onto client-owned cloud hardware. Eliminates vendor lock-in, cuts monthly AWS/PaaS hosting bills by ~80%, and guarantees complete data sovereignty.

1. PRODUCTION INFRASTRUCTURE ARCHITECTURE



• Dedicated NVMe I/O Performance

Unlike shared PaaS containers that throttle CPU cycles, a dedicated Linux VPS (e.g. Hetzner CX22) provides dedicated vCPU execution and NVMe disk speeds for sub-50ms TTFB.

• Isolated Private Docker Network

Database (PostgreSQL/MySQL) and Redis instances are bound strictly to internal Docker bridge sockets. Database ports (5432 / 3306) are never exposed to the public internet.

2. THE 7-STAGE PRODUCTION HARDENING PROTOCOL

- ✓ **Stage 1: Base Linux OS Hardening & User Privilege:** Clean Ubuntu 22.04/24.04 LTS provisioning, automated security patching (unattended-upgrades), non-root execution, and hardened SSH configuration.
- ✓ **Stage 2: UFW Firewall & Network Lockdown:** Default-deny incoming traffic policy. Explicitly white-lists ports 22 (SSH), 80 (HTTP), and 443 (HTTPS). Blocks all direct external access to databases, Redis, and internal telemetry.
- ✓ **Stage 3: Containerized PaaS Engine (Coolify + Docker):** Automated installation of isolated Docker daemon, Traefik edge reverse proxy, and self-hosted control plane for zero-overhead resource management.
- ✓ **Stage 4: Automated Let's Encrypt SSL & HTTP/2 Delivery:** Cryptographic verification via ACME challenges. Full HTTP/2 and TLS 1.3 encryption with automated 60-day auto-renewal daemons.
- ✓ **Stage 5: Persistent Volume Architecture & Disk Pruning:** Storage isolation via named Docker volumes preserving uploads across rebuilds. Weekly automated Docker pruning cron eliminating dangling images and cache bloat.
- ✓ **Stage 6: Zero-Downtime Git Push-to-Deploy Webhooks:** Direct repository integration via secure deploy keys. Pushes to production branches trigger automated container builds and atomic swaps with zero downtime.
- ✓ **Stage 7: Laravel Framework Production Tuning:** Execution of production optimization routines: configuration caching, route caching, event discovery, storage symlinking, and OPcache preloading.

3. PRODUCTION OPTIMIZATION COMMANDS

Every deployed container is optimized for maximum throughput and minimal memory overhead during build:

```
# Production Framework Optimization Pipeline
php artisan config:cache php artisan route:cache php artisan view:cache
php artisan event:cache php artisan storage:link php artisan migrate --force
# Automated Host Maintenance (Weekly Disk Health Cron)
0 3 * * 0 docker system prune -af --volumes
```

4. AUTOMATED OFF-SITE DISASTER RECOVERY (PREMIUM)

Production databases are never left vulnerable on a single physical host. In the Premium architecture, automated snapshot routines are configured to ship encrypted database dumps to S3-compatible cloud storage:

• **Daily Scheduled DB Dumps**

Automated database backup jobs execute every 24 hours during off-peak hours (03:00 UTC), gzipping SQL dumps directly from the container.

• **Off-Site Cloudflare R2 / AWS S3 Storage**

Dumps are pushed immediately to private, versioned bucket storage. In the event of catastrophic server hardware failure, full state can be restored in minutes.

5. CLIENT SECURITY & HANDOVER PROTOCOL

To eliminate developer dependency and guarantee client security, every completed deployment follows a strict 3-step security handover checklist:

SECURITY STEP	CLIENT ACTION REQUIRED UPON DELIVERY	SECURITY BENEFIT
1. Root Password Rotation	Client logs in via SSH and executes passwd to set a private master root password.	Ensures only the business owner holds master server credentials.
2. Temporary SSH Key Revocation	Remove contractor SSH public keys from ~/.ssh/authorized_keys with one command.	Eliminates ongoing external SSH access; locks down host access completely.
3. Coolify Master Admin Password	Log into the web dashboard and rotate the administrator email and master password.	Grants 100% control over applications, container logs, and deployment webhooks.

6. PACKAGE DELIVERABLES MATRIX

SCOPE ITEM	BASIC (\$45)	STANDARD (\$95)	PREMIUM (\$175)
SSL Certificate Repair / Nginx Configuration Fix	✓ Included	✓ Included	✓ Included
Clean Linux OS Provisioning & Docker Engine Setup	—	✓ Included	✓ Included
Coolify Self-Hosted Control Plane Installation	—	✓ Included	✓ Included
Full Laravel Application Deployment (1 Live Domain)	—	✓ Included	✓ Included
Isolated PostgreSQL / MySQL Container Database	—	✓ Included	✓ Included
UFW Host Firewall Lockdown & Port Security	—	✓ Included	✓ Included
Automated Git Push-to-Deploy Webhooks	—	✓ Included	✓ Included
Redis Cache & Asynchronous Queue Worker Setup	—	—	✓ Included
Automated Daily S3 / Cloudflare R2 Database Backups	—	—	✓ Included
Written Architecture Handover & Security Runbook	Optional Extra	✓ Included	✓ Included